LIMITS OF NONEMPTY QC SYSTEMS ARE NONEMPTY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1196-1203, following the statement at 1188-1194.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Existence of directed inverse limits. lemma-directed-inverse-system-has-limit (same chapter, card L26291, cited at ~1190-1192). The limit exists (hypothesis of the lemma).
- Scheme-level limit nonemptyness. Limits, Lemma limits-lemma-limit-nonempty (cross-chapter, cited at ~1201-1202). U = lim U_i is nonempty for nonempty affine U_i.

Source reference. Original lemma 07SF: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 878–878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 086W: focused-seeds/geometry/S9-spaces-limits-limit-nonempty/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $X = \lim_{i \in I} X_i$ be the limit of a directed inverse system of algebraic spaces over S with affine transition morphisms (Lemma 07SF). If each X_i is quasi-compact and nonempty, then $|X|$ is nonempty.*

Proof. Choose $0 \in I$. Choose an affine scheme U_0 and a surjective étale morphism $U_0 \to X_0$. Set $U_i = X_i \times_{X_0} U_0$ and $U = X \times_{X_0} U_0$. Then each U_i is a nonempty affine scheme. Hence $U = \lim U_i$ is nonempty (Limits, Lemma 086W) and thus X is nonempty. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.